

Optional Notes on Euclid's Lemma and Basis Representation Theorem

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1 Lemma for Basis Representation Theorem

$$n < B^n \text{ whenever } B \geq 2$$

2 Proof of Lemma

$$n < B^n$$

Because $2^n \leq B^n$ for any $B \geq 2$ then if we can prove $n < 2^n$ that is sufficient.

For example consider $4 < 2^4$,

We can show that as follows:

$$\underbrace{1+1+1+1}_{\text{n terms}} < \underbrace{2 \cdot 2 \cdot 2 \cdot 2}_{\text{n terms}}$$

$$\underbrace{1+1+1+1}_{\text{n terms}} < \underbrace{1+1}_{2} + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1$$

$$\qquad \qquad \qquad \underbrace{\hspace{10em}}_{2 \cdot 2}$$

$$\qquad \qquad \qquad \underbrace{\hspace{18em}}_{2 \cdot 2 \cdot 2}$$

$$\qquad \qquad \qquad \underbrace{\hspace{26em}}_{2 \cdot 2 \cdot 2 \cdot 2}$$

Because the expansion $2 \cdot 2 \cdot 2 \cdot 2$ above has 4 vertical levels, we know that there at least 4 ones in the series of additions.

Therefore, $4 < 2^4$ and in general $n < B^n$ if $B \geq 2$ by considering similar expansions of additions of ones.

3 Proof 2 of Lemma

$$n = \underbrace{1 + 1 + 1 + 1}_{n \text{ terms}} \leq 1 + B + B^2 + \cdots + B^{n-1}$$

since $B \geq 2$

$$1 + B + B^2 + \cdots + B^{n-1} = \sum_{i=0}^{n-1} B^i = \frac{B^n - 1}{B - 1}$$

Using the equation for $\sum_{i=0}^{n-1} B^i$ from the induction lecture.

$$\frac{B^n - 1}{B - 1} \leq \frac{B^n - 1}{2 - 1} = B^n - 1$$

since $B \geq 2$

$$B^n - 1 < B^n$$

Therefore, $n < B^n$

(from Andrews)

4 Proof 2 of Basis Representation Theorem

We need to show there is a unique set of numbers $C_1, C_2, \dots, C_{N-2}, C_{N-1}$ with all $0 \leq C_i < B$ that satisfy $A = C_1 \cdot B^N + C_2 \cdot B^{N-1} + \cdots + C_{N-2} \cdot B + C_{N-1}$

Given any representation $A = C_1 \cdot B^N + C_2 \cdot B^{N-1} + \cdots + C_{N-2} \cdot B + C_{N-1}$, if C_{N-1-M} is the first non zero C_i from the right, then we can rewrite A as $C_1 \cdot B^N + C_2 \cdot B^{N-1} + \cdots + C_{N-1-M} \cdot B^M$

This means that

$$\begin{aligned}
A - 1 &= C_1 \cdot B^N + C_2 \cdot B^{N-1} + \dots + C_{N-1-M} \cdot B^M - 1 = \\
&C_1 \cdot B^N + C_2 \cdot B^{N-1} + \dots + (C_{N-1-M} - 1)B^M + (B^M - 1) = \\
&C_1 \cdot B^N + C_2 \cdot B^{N-1} + \dots + (C_{N-1-M} - 1)B^M + (B^M - B^{M-1}) \\
&\quad + (B^{M-1} - B^{M-2}) \\
&\quad \vdots \\
&\quad + (B^2 - B) \\
&\quad + (B - 1) \\
&= C_1 \cdot B^N + C_2 \cdot B^{N-1} + \dots + (C_{N-1-M} - 1)B^M + (B - 1) \cdot B^{M-1} \\
&\quad + (B - 1) \cdot B^{M-2} \\
&\quad \vdots \\
&\quad + (B - 1) \cdot B^1 \\
&\quad + (B - 1) \cdot B^0
\end{aligned}$$

Since $0 \leq B - 1 < B$, this means there exists a representation for $A - 1$ with each coefficient greater than or equal to zero and less than B .

We can write the number of representations of a number K in base B as $R_B(K)$

We showed above that for each representation of A , we can derive a new representation of $A - 1$, which means $R_B(A - 1) \geq R_B(A)$

This also means that $R_B(A) \geq R_B(A + 1)$, as we can use the same idea for each representation of $A + 1$

Then we see

$$\begin{aligned}
R_B(1) &\geq \dots \geq R_B(A - 2) \geq R_B(A - 1) \geq R_B(A) \geq \\
R_B(A + 1) &\geq R_B(A + 2) \geq \dots
\end{aligned}$$

Using the lemma from the beginning of this lecture, $B^A > A$, when $B \geq 2$ we can continue

$$\begin{aligned} R_B(1) &\geq \dots \geq R_B(A-2) \geq R_B(A-1) \geq R_B(A) \geq \\ R_B(A+1) &\geq R_B(A+2) \geq \dots \geq R_B(B^A) \end{aligned}$$

It is easy to show that B^A has at least one representation

$$\text{So } R_B(B^A) \geq 1$$

$$1 \cdot B^A + 0 \cdot B^{A-1} + 0 \cdot B^{A-2} + \dots + 0 \cdot B^1 + 0$$

Then we have

$$\begin{aligned} R_B(1) &\geq \dots \geq R_B(A-2) \geq R_B(A-1) \geq R_B(A) \geq \\ R_B(A+1) &\geq R_B(A+2) \geq \dots \geq R_B(B^A) \geq 1 \end{aligned}$$

We also know that $R_B(1) = 1$, since the only representation of 1 is 1, because any higher order terms $C_i \cdot B^{N+1-i}$ would make for a number larger than 1 as $B \geq 2$

Then we have

$$\begin{aligned} 1 &\geq R_B(1) \geq \dots \geq R_B(A-2) \geq R_B(A-1) \geq R_B(A) \geq \\ R_B(A+1) &\geq R_B(A+2) \geq \dots \geq R_B(B^A) \geq 1 \end{aligned}$$

Since $1 \geq R_B(A)$ and $R_B(A) \geq 1$, we know $R_B(A) = 1$ which means there is a unique representation of A in base B .